

**ZHESTKOV
UNIVERSITY**

МАТАНАЛИЗ

**Замена переменных в
кратных интегралах**

Мы уже говорили о том, как искать двойной, тройной и даже n -кратный интеграл

Рецепт: свести к повторному, оптимизировав порядок интегр.

Но это мало. Вспомним опр. интегр.

$$\int_0^{\pi/4} \tan x \, dx = \int_0^{\pi/4} \frac{\sin x}{\cos x} \, dx = \left| \begin{array}{l} \cos x = t \\ dt = -\sin x \, dx \\ x=0 \rightarrow t=1 \\ x=1 \rightarrow t=\frac{1}{\sqrt{2}} \end{array} \right| = \int_1^{\frac{1}{\sqrt{2}}} -\frac{dt}{t} = \int_{\frac{1}{\sqrt{2}}}^1 \frac{dt}{t} = \ln t \Big|_{\frac{1}{\sqrt{2}}}^1 = \ln \sqrt{2}$$

Пусть $x = \varphi(t)$ неупр. диффр. на $[a, b]$, f неупр. на $\varphi([a, b])$, тогда

$$\int_a^b f(\varphi(t)) \varphi'(t) \, dt = \int_{\varphi(a)}^{\varphi(b)} f(x) \, dx$$

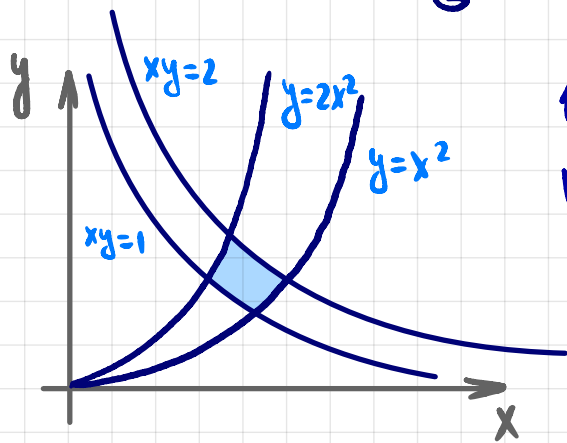
Замена была важной интегр. Так будем и в краевых $\left. \begin{array}{l} x = x(u, v) \\ y = y(u, v) \end{array} \right\}$ для двойных

Пусть $F: \mathbb{R}_{uv}^2 \supset G \xrightarrow[\text{откр. изм.}]{\text{Ф-т}} G \subset \mathbb{R}_{xy}^2$, F н.д. на $\bar{G}$, $\frac{\partial(x, y)}{\partial(u, v)} \neq 0 \, \forall (u, v) \in G$

Пусть f неупр. диффр. на $\bar{G}$, тогда

$$\iint_G f(x, y) \, dx \, dy = \iint_G f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, du \, dv$$

Пр.1 Найдите $\iint_G y^3 dx dy = I$; $G: \{y=x^2, y=2x^2, xy=1, xy=2\}$



$$u = \frac{y}{x^2}, \quad u=1, u=2$$

$$v = xy, \quad v=1, v=2$$

$$\tilde{G}: \{1 \leq u \leq 2, 1 \leq v \leq 2\}$$

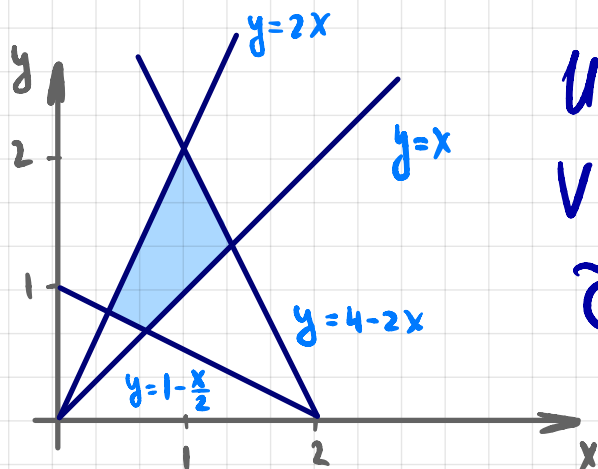
Найдем якобиан $\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = J$

$$x = u^{-\frac{1}{3}} v^{\frac{1}{3}}, \quad y = u^{\frac{1}{3}} v^{\frac{2}{3}}$$

$$J = \begin{vmatrix} -\frac{1}{3} u^{-\frac{4}{3}} v^{\frac{1}{3}} & \frac{1}{3} u^{-\frac{1}{3}} v^{-\frac{2}{3}} \\ \frac{1}{3} u^{-\frac{2}{3}} v^{\frac{2}{3}} & \frac{2}{3} u^{\frac{1}{3}} v^{-\frac{1}{3}} \end{vmatrix} = -\frac{1}{3u}$$

$$I = \iint_{\tilde{G}} \cancel{u} v^2 \cdot \frac{1}{3\cancel{u}} du dv = \frac{1}{3} \int_1^2 du \int_1^2 v^2 dv = \frac{7}{9}$$

Пр.2. Найдите $\iint_G \frac{1}{y} dx dy = I$; $G: \{y=x, y=2x, y=1-\frac{x}{2}, y=4-2x\}$
 $u=1 \quad u=2 \quad v=\frac{1}{2} \quad v=2$



$$u = \frac{y}{x}, \quad u \in [1, 2]$$

$$v = \frac{y}{2-x}, \quad v \in [\frac{1}{2}, 2]$$

$\tilde{G}$ - параллелограмм!

$$x = \frac{2v}{u+v}$$

$$y = \frac{2uv}{u+v}$$

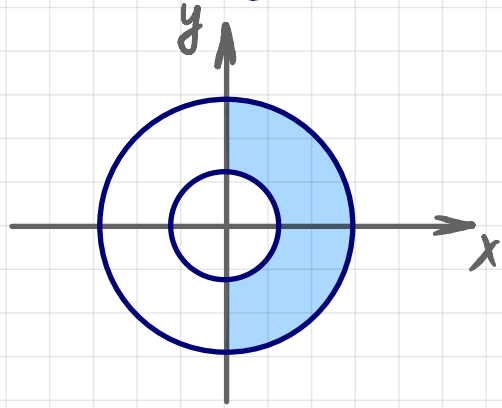
$$|J| = \frac{4uv}{(u+v)^3}$$

$$I = \iint_G \frac{1}{y} dx dy = \iint_{\tilde{G}} \frac{\cancel{u+v}}{2\cancel{u+v}} \cdot \frac{4\cancel{u}\cancel{v}}{(u+v)^3} du dv =$$

$$= 2 \int_1^2 du \int_{\frac{1}{2}}^2 \frac{1}{(u+v)^2} dv = 2 \ln \frac{5}{4}$$

Часто удобно использ. полярные коорг. $\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases} \quad r = |z| = r > 0$

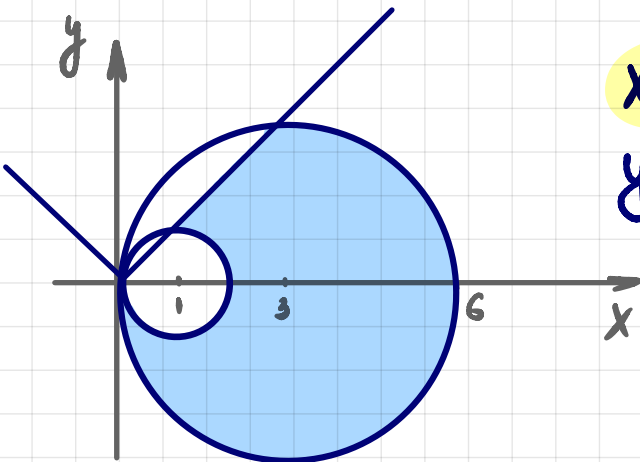
Пр. 3. $\iint_G x dx dy = I$ $G: \{1 \leq x^2 + y^2 \leq 4, x > 0\}$



$$\begin{aligned} x &= r \cos \varphi \\ y &= r \sin \varphi \end{aligned} \quad \tilde{G}: \begin{aligned} 1 &\leq r \leq 2 \\ -\frac{\pi}{2} &< \varphi < \frac{\pi}{2} \end{aligned}$$

$$I = \iint_{\tilde{G}} r \cos \varphi \cdot r dr d\varphi = \int_1^2 r^2 dr \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \varphi d\varphi = \left(\frac{8}{3} - \frac{1}{3}\right) \cdot 2 = \frac{14}{3}$$

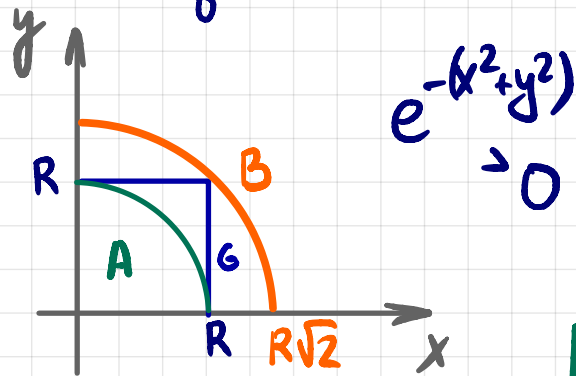
Пр. 4 $\iint_G x dx dy = I$. $G: \{2x \leq x^2 + y^2 \leq 6x, y \leq |x|\}$



$$\begin{aligned} x &= r \cos \varphi \\ y &= r \sin \varphi \end{aligned} \quad \begin{aligned} \varphi &: [-\frac{\pi}{2}, \frac{\pi}{4}] \\ r &: [2 \cos \varphi, 6 \cos \varphi] \end{aligned}$$

$$I = \iint_{\tilde{G}} r \cos \varphi \cdot r dr d\varphi = \int_{-\frac{\pi}{2}}^{\frac{\pi}{4}} \cos \varphi d\varphi \int_{2 \cos \varphi}^{6 \cos \varphi} r^2 dr = \frac{13}{6} (9\pi + 8)$$

Зр. 5. $\int_0^{+\infty} e^{-x^2} dx$



$$\iint_A e^{-(x^2+y^2)} dx dy \leq \iint_G e^{-(x^2+y^2)} dx dy = \iint_B e^{-(x^2+y^2)} dx dy \quad (*)$$

$$\iint_G e^{-(x^2+y^2)} dx dy = \int_0^R e^{-x^2} dx \int_0^R e^{-y^2} dy = \left(\int_0^R e^{-x^2} dx \right)^2$$

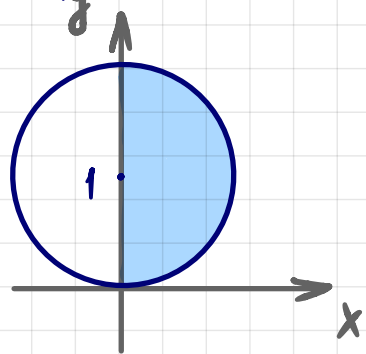
В змінній внутр. перетв. к поляр.

$$\iint_A e^{-(x^2+y^2)} dx dy = \int_0^{\frac{\pi}{2}} d\varphi \cdot \int_0^R e^{-r^2} \cdot r \cdot dr = \frac{\pi}{4} (1 - e^{-R^2})$$

$$(*) \quad \frac{\pi}{4} (1 - e^{-R^2}) \leq \left(\int_0^R e^{-x^2} dx \right)^2 \leq \frac{\pi}{4} (1 - e^{-2R^2}) \Rightarrow \int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

$\downarrow R \rightarrow \infty$
 $\frac{\pi}{4}$

Зр. 6. Знайти $\iint_G xy dx dy$; $G: \{x^2 + y^2 = 2y, x > 0\}$



$$\begin{aligned} x &= r \cos \varphi \\ y &= 1 + r \sin \varphi \\ r &= r \end{aligned}$$

$$\begin{aligned} \varphi &: [-\frac{\pi}{2}, \frac{\pi}{2}] \\ r &: [0, 1] \end{aligned}$$

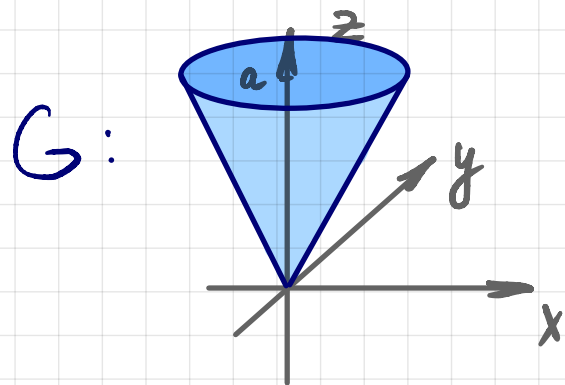
$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\varphi \int_0^1 r \cos \varphi (1 + r \sin \varphi) \cdot r dr = \frac{2}{3}$$

$$\begin{aligned} \sqrt{\frac{x^2}{a^2}} + \sqrt{\frac{y^2}{b^2}} &= 1 \\ x &= a \cos \varphi \\ y &= b \sin \varphi \\ r &= \dots \end{aligned}$$

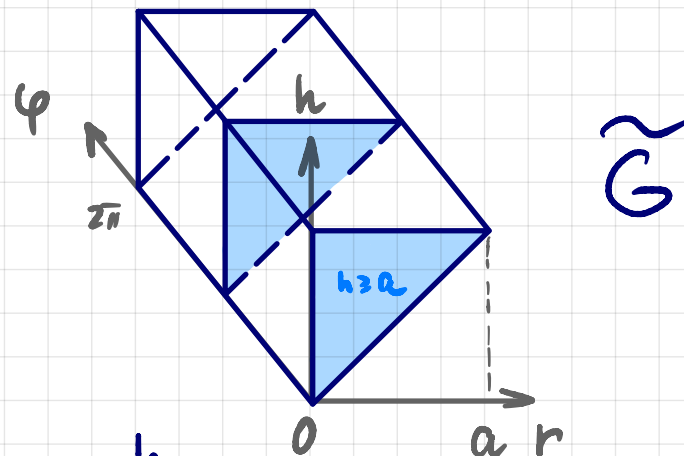
Для тройных интегралов тоже есть станд. замены

Цилиндрические коорд. $\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \\ z = h \end{cases} \quad \begin{vmatrix} \cos \varphi & -r \sin \varphi & 0 \\ \sin \varphi & r \cos \varphi & 0 \\ 0 & 0 & 1 \end{vmatrix} = J = r > 0$

Пр. 7. $\iiint_G \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + z^2}} dx dy dz \quad G: \{\sqrt{x^2 + y^2} \leq z \leq a\}$

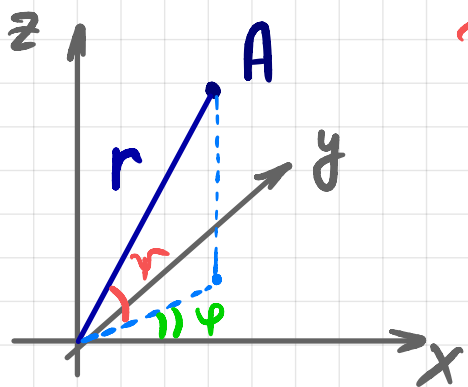


$$\begin{aligned} x &= r \cos \varphi \\ y &= r \sin \varphi \\ z &= h \end{aligned} \quad \begin{aligned} r &\leq h \leq a \\ \varphi &\in [0, 2\pi] \end{aligned}$$



$$\iiint_G \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + z^2}} dx dy dz = \iiint_{\tilde{G}} \frac{r^2}{\sqrt{r^2 + h^2}} dr d\varphi dh = \int_0^{2\pi} d\varphi \int_0^a dh \int_0^h \frac{r^2}{\sqrt{r^2 + h^2}} dr = \frac{\pi}{6} (2 - \sqrt{2}) a^4$$

Сферические координаты



ψ -широта $\psi: [-\frac{\pi}{2}, \frac{\pi}{2}]$

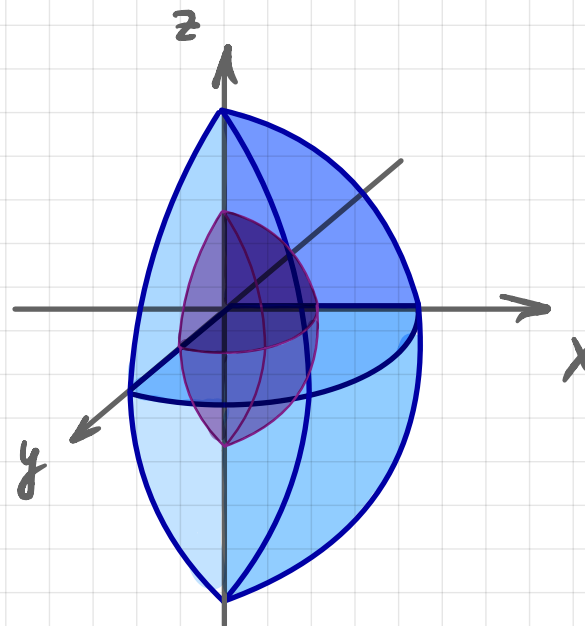
φ -долгота $\varphi: [-\pi, \pi]$

$$x = r \cos \psi \cos \varphi$$

$$y = r \cos \psi \sin \varphi$$

$$z = r \sin \psi$$

$$J = r^2 \cos \psi > 0$$



Пр. 8 $\iiint_G (x^2 + y^2 - z^2) dx dy dz$ $G: \{1 \leq x^2 + y^2 + z^2 \leq 4, x > 0, y > 0\}$

$\tilde{G}: \psi: [-\frac{\pi}{2}, \frac{\pi}{2}]$

$\varphi: [0, \frac{\pi}{2}]$

$r: [1, 2]$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\psi \int_0^{\frac{\pi}{2}} d\varphi \int_1^2 (r^2 \cos^2 \psi - r^2 \sin^2 \psi) r^2 \cos \psi dr =$$

$$= \int_0^{\frac{\pi}{2}} d\varphi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos 2\psi \cos \psi d\psi \int_1^2 r^4 dr = \frac{31\pi}{15}$$

$\sin \psi = t$

Геометр. и физ. прилож. криволинейных интегр.

Площадь G : $\iint_G 1 \cdot dx \cdot dy$

Объем G : $\iiint_G 1 \cdot dx \cdot dy \cdot dz$

Здесь есть физич. прилож.

Пр. 9. Найдите ц.м. полушара

$$G: \{x^2 + y^2 + z^2 \leq R^2, z \geq 0\}$$

$$\rho = \frac{\rho_0}{\sqrt{x^2 + y^2 + z^2}} \quad C(x_c, y_c, z_c)$$

$x_c = y_c = 0$ (в центре сферы)

$$x_c = \frac{M_x}{M}; \quad y_c = \frac{M_y}{M}; \quad z_c = \frac{M_z}{M}$$

$$M = \iiint_G \rho \, dx \, dy \, dz \quad \text{Масса}$$

$$M_x = \iiint_G x \cdot \rho \, dx \, dy \, dz \quad \text{Статический момент}$$

$$M = \iiint_G \frac{\rho_0}{\sqrt{x^2 + y^2 + z^2}} \, dx \, dy \, dz = \rho_0 \int_0^{\frac{\pi}{2}} dr \int_0^{2\pi} d\varphi \int_0^R \frac{1}{r} \cdot r^2 \cos r \, dr = \pi \rho_0 R^2$$

Аналог. M_z, M_x^0, M_y^0

Отв: $(\frac{R}{3}, 0, 0)$

$$\frac{M_z}{M} = \frac{R}{3}$$

